# Paraminder Dhiman

## Technical Artist · Tools and Pipelines

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## SUMMARY

Technical Artist with 7 years at Ubisoft, including the shipped AAA title Skull and Bones (PS5, Xbox Series X|S, PC). I build artist-facing tools and asset pipelines in MaxScript and Python across 3ds Max, Houdini and Ubisoft's in-house Anvil engine. My work also covers technical animation, cinematics and technical design, and outside the studio I edit video and run a self-hosted server platform for 20 users.

## EXPERIENCE

**Ubisoft India Studios** — Pune, India
**Technical Artist · Skull and Bones** — Feb 2023 – Present
*Ship building and ship customisation pipeline*

- Built an ornament placement tool in 3ds Max (MaxScript) that replaced a manual Houdini workflow, cutting placement from up to 1.5 hours to about 15 minutes per ship across 22+ ships.
- Solved asset mirroring for an engine that does not accept negative scale by implementing quaternion-based mirroring, so ornaments mirror correctly across the hull.
- Rolled the placement tool into the studio pipeline, where it is used by art teams in local and overseas studios.
- Took over a broken, inherited sail animation pipeline (SSDR cloth-to-bone baking in Houdini), reverse-engineered how it works and became its sole owner on the project; now repairing it.
- Built a new ship spawning system on the engine's entity-building tool to assemble and spawn ship meshes.
- Optimised Houdini ship-building workflows to speed up artist iteration.
- Profiled and optimised shaders and materials to keep ship content within performance budgets.
- Mentored junior technical artists in MaxScript and Python scripting, Houdini and Anvil workflows.

**Ubisoft India Studios** — Mumbai, India
**Junior Technical Artist · Skull and Bones** — Oct 2021 – Feb 2023
*Pets (player vanity items)*

- Delivered 50+ pets, supporting each one from concept art through ready-to-animate (RTA) setup to final integration in the engine.
- Rebuilt the pet integration pipeline and the pet spawning system.
- Restructured and rebuilt the full pet animation graph in the in-house animation graph tool.
- Worked across 3ds Max, Blender, Maya and the Anvil engine.

**Ubisoft India Studios** — Mumbai, India
**Junior Technical Level Designer (Cinematics) · Prince of Persia: The Sands of Time Remake** — Nov 2019 – Oct 2021
*Original Ubisoft Mumbai and Pune version (unreleased)*

- Integrated and validated character animations for cutscenes in the in-house timeline-based cinematics editor, working with character, audio and lighting tracks.
- Lit around 20 cutscenes.
- Owned and resolved all cinematics-related bugs.
- Prepared animation data in MotionBuilder for cutscenes, with cleanup where needed.
- Scripted cinematic triggers.
- Edited cinematic video in Adobe Premiere Pro.
- Maintained cutscene documentation for the team.

## PROJECTS

**Black Hole · Blender and Unreal** 2023 – 2026

*Personal · Blender, Python, OSL · Unreal Engine 5, HLSL*

- **Black Hole Studio** is a Blender extension that renders a black hole by bending light the way general relativity does. Fast mode bakes every light path into a lookup image so it runs on the GPU and in Eevee; Exact mode traces each ray in an OSL shader in Cycles. I designed it and wrote much of the code, with parts developed with AI assistance. Code on GitHub.
- **Black Hole Shader** is a real-time Unreal Engine 5 material with a custom HLSL node: light bent by the Schwarzschild equations, a volumetric accretion disk, Doppler beaming and blackbody colour. Version 2.0 made the bending physically accurate, with artist controls on top. Free on Fab.

**Blender Extensions** 2026

*Personal · Blender, Geometry Nodes, Python*

- **Unravel** tears cloth: paint where it should tear, and it cuts the holes, thickens their edges and grows frayed threads, built in Geometry Nodes with a Python add-on. Published on Blender Extensions.
- **UV Island Colorizer** gives every UV island its own colour and bakes a colour ID map, across many meshes at once. Published on Blender Extensions.
- **Positive Mirror** mirrors objects with a positive scale instead of −1, so normals and exports stay correct. Code on GitHub.

**Home Server and Automation** 2026

*Personal · Unraid, Docker, Python*

- Designed and run a self-hosted services platform for 20 users, with Docker containers, secure remote access (Tailscale, Cloudflare Tunnels), network-wide DNS and DHCP, and scheduled backups.
- Wrote a Python Discord bot (discord.py) that automates user onboarding and account provisioning, deployed as a container and versioned in Git.

**More projects**

Every project, with how and why each was built, is on my portfolio at paraminder-dhiman.github.io.

## SKILLS

**Scripting and programming**
MaxScript, Python, C#, HLSL, C++ (working knowledge), VEX (working knowledge)

**Technical animation**
Animation graphs, Cloth-to-bone baking (SSDR), Animation data preparation in MotionBuilder

**DCC software**
3ds Max, Houdini, Blender, Maya, MotionBuilder

**Video**
DaVinci Resolve (editing, Fusion, colour grading), Adobe Premiere Pro

**Technical art**
Tool development, Asset integration pipelines (RTA), Shader and performance profiling

**Cinematics and technical design**
Timeline-based cutscene assembly, Cutscene lighting, Cinematic trigger scripting

**Engines and pipeline**
Anvil (Ubisoft in-house), Unreal Engine 5, Perforce, Git

**Infrastructure**
Docker, Unraid, Home networking and remote access

## EDUCATION

**Abertay University** Professional Masters in Games Development (MProf)
Dundee, UK • 2019

**Supinfogame Rubika** Bachelor cycle in Video Game, Game Design specialisation (3 years, 180 ECTS)
Pune, India • 2015 – 2018